

QRP VLDO V2

M9OMS

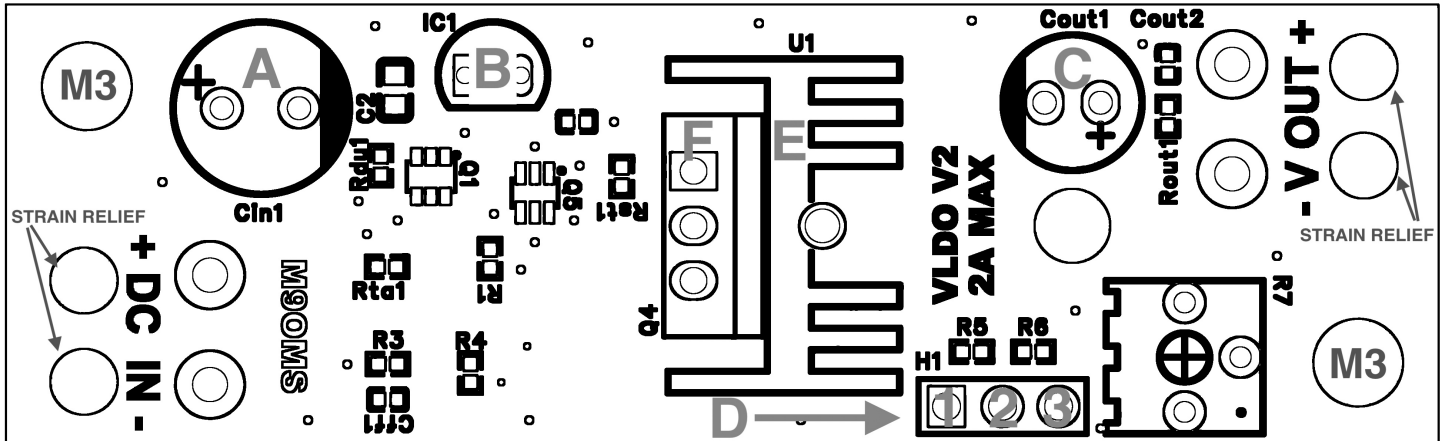


Figure 1: Top of board, with assembly steps labelled A–F.

Assembly instructions

1. **A** — Solder Cin1 (470 μ F, red electrolytic capacitor). Observe polarity — the long lead is positive.
2. **B** — Solder IC1 (78L05, TO-92). Take care not to bridge the pads.
3. **C** — Solder Cout1 (220 μ F, silver polymer capacitor). Observe polarity.
4. **D** — Solder H1 (header pins). Tack one leg down first, then reflow that joint while pressing the header flat against the board to get it square. Watch your fingers — the pins get hot. Once it sits flat and square, solder the remaining legs.

⚠ **HANDLE PMOS & VREF WITH CARE — ESD-sensitive devices.** ⚠

Decide now how you will mount the PMOS: with the included heat sink (continue to step **E**), or bottom-mounted directly to an external case or cooling solution (skip to step **G**). **The PMOS is fully isolated — no isolating pads are necessary.** Using an isolating pad will increase thermal resistance. Use thermal paste only.

Apply a small amount of thermal paste to the back of the PMOS. Place it onto the heat sink and gently secure it with the M3 machine screw. Do not tighten fully yet — the PMOS should still be able to move from side to side, so you can ease its legs into the board pads in step **F**.

5. **E** — Place the PMOS + heat sink assembly onto the top of the board. The heat sink should sit flush. Solder the heat sink pin: this joint is mechanical only and carries no current, but apply enough solder to immobilise the pin. Use a HOT iron for this. Ensure the assembly is secure, straight and square before continuing.
6. **F** — Solder the PMOS gate pad (the square one) at normal iron temperature. Then solder the two round pads — additional heat will be required, particularly for the centre pin. Once all three pins are soldered, tighten the M3 screw fully.
7. **G** — *Bottom-mount option (PMOS bonded directly to your case or external cooling solution).* Dry-fit the PMOS referencing the clearance hole on the rear of the board, with its writing facing the PCB, and note where the legs need to bend. Bend them carefully to a right angle. Bond the PMOS to your cooling solution first (don't forget thermal paste), then fit the board over the legs and solder them — solder after everything is

mechanically secured for the best result. The clearance hole stays accessible from the top of the board, so you can tighten or remove the PMOS mounting screw later without desoldering.

Wiring and strain relief

The input and output pads accommodate up to 18 AWG stranded wire – keep the cables short, pay close attention to polarity. Use the strain relief holes to secure your cables:

- Pass the wire through the strain relief hole from the top of the board.
- Bring it back down through the pad hole from the bottom, leaving a tight loop.
- Pull the loop snug, then solder. Any pull on the cable now acts on the loop rather than the solder joint.

A small cable tie through the strain relief holes can work as an alternative.

Selecting the output voltage

The header pins (H1) allow you to select the desired output voltage. Jumpers/shunts are included.

- **No shunt** $\simeq 9\text{ V}$
- **Shunt pins 1 + 2** $\simeq 12\text{ V}$
- **Shunt pins 1 + 3** $\simeq 13.8\text{ V}$

Check the output voltage is correct before connecting the board to sensitive equipment. Adjust the output (with $V_{in} > V_{out}$) using the trim potentiometer R7. If there is any subsequent drift, re-adjust R7.

Recommended operating parameters

- **Input:** 8*–18 V DC
- **Regulated output:** 9.0, 12.0, 13.8 V DC (trimmable to $\pm 20\text{ mV}$ at the chosen setpoint)
- **Max current:** 2 A

* A DC supply stiff enough to keep the input voltage above this threshold is necessary.

Heat sink and housing

The supplied heat sink should dissipate up to 3 W of heat at a 50% duty cycle, provided it has sufficient ventilation.

There are various ways to house the board. Please ensure there are plenty of ventilation holes when housing it in a project box or 3D-printed case.

For larger current draws and/or voltage drops, it is advisable to house the board in a metal project box, with the PMOS bonded directly to the case for better heat dissipation (see step G). The Multicomp MC003550 has good dimensions and is available from CPC for under £4.

Board layout note

The bottom of the circuit board is a +DC OUT power plane (for thermal management). The top of the board is a ground plane. Care must be taken not to short the two sides — for example, when securing the board down. Use plastic or cardboard washers with metal hardware.